Enxin Song

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Research Overview

My research centers on multimodal models and agents, with key areas of focus including:

- **Efficient Long-Sequence Modeling**, scaling models to long and streaming inputs such as video with sparse attention, memory, token compression, and recurrent architectures.
- **Real-Time Interactive Agents**, building systems that perceive continuous streams and decide when and how to respond.
- **Benchmarking and Evaluation**, building benchmarks and evaluation toolkits that pose challenging real-world tasks, probe the boundaries of model capabilities, and provide insights for future improvement.

Education

Ph.D. Fall 2026 – Present	University of Pennsylvania Advisor: Jiatao Gu
M.S. CS 2023-2026	Zhejiang University Advisor: Gaoang Wang
Visiting Spring/Summer 2025	University of California San Diego Advisor: Zhuowen Tu
B.S. 2019-2023	Dalian University of Technology Software Engineering

Employment

Research Intern 2023-2024	Microsoft Research Asia Working on Video Understanding Mentor: Xun Guo

Teaching

CIS 6280 World Models (Graduate) Teaching Assistant, University of Pennsylvania Instructor: Prof. Jiatao Gu	Fall 2026
ECE 445 Senior Design (Undergraduate) Teaching Assistant, Zhejiang University - University of Illinois Urbana-Champaign Instructor: Prof. Gaoang Wang	Spring 2024

Selected Publications

The * sign denotes equal contribution.

Peer-Reviewed Papers

C7 Ruizhe Chen, Zhiting Fan, Yang Shi, Enxin Song, Wenhao Chai, Tongkun Guan, Songtao Jiang, Ruilin Luo, Yuanxing Zhang, Shuai Bai, Sibo Song, Zhibo Yang, and Zuozhu Liu. Video* Bench: Benchmarking Fine-grained Video Perception with "Think with Videos". In *ACM International Conference on Multimedia (MM)*. 2026.

C6 Enxin Song*, Wenhao Chai, Shusheng Yang, Ethan J. Armand, Xiaojun Shan, Haiyang Xu, Jianwen Xie, and Zhuowen Tu. VideoNSA: Native Sparse Attention Scales Video Understanding. In *ICLR*. 2026.

J1 Enxin Song*, Wenhao Chai*, Tian Ye, Jenq-Neng Hwang, Xi Li, and Gaoang Wang. MovieChat+: Question-aware Sparse Memory for Long Video Question Answering. In *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*. 2025.

C5 Haodong Duan, Xinyu Fang, Junming Yang, Xiangyu Zhao, Yuxuan Qiao, Mo Li, Amit Agarwal, Zhe Chen, Lin Chen, Yuan Liu, Yubo Ma, Hailong Sun, Yifan Zhang, Shiyin Lu, Tack Hwa Wong, Weiyun Wang, Peiheng Zhou, Xiaozhe Li, Chaoyou Fu, Junbo Cui, Jixuan Chen, Enxin Song, Song Mao, Shengyuan Ding, Tianhao Liang, Zicheng Zhang, Xiaoyi Dong, Yuhang Zang, Pan Zhang, Jiaqi Wang, Dahua Lin, Kai Chen. Vlmevalkit: An open-source toolkit for evaluating large multi-modality models. In *ACM international conference on multimedia (MM)*. 32nd.

C4 Weili Xu*, Enxin Song*, Wenhao Chai*, Tian Ye, and Gaoang Wang. Bringing RNNs Back to Efficient Open-Ended Video Understanding. In *International Conference on Computer Vision, (ICCV)*. 2025.

C3 Wenhao Chai*, Enxin Song*, Yilun Du, Chenlin Meng, Vashisht Madhavan, Omer Bar-Tal, Jenq-Neng Hwang, Saining Xie, and Christopher D. Manning. AuroraCap: Efficient, Performant Video Detailed Captioning and a New Benchmark. In *International Conference on Learning Representations (ICLR)*. 2025.

C2 Jinbin Bai, Tian Ye, Wei Chow, Enxin Song, Qing-Guo Chen, Xiangtai Li, Zhen Dong, Lei Zhu, Shuicheng Yan. Meissonic: Revitalizing Masked Generative Transformers for Efficient High-Resolution Text-to-Image Synthesis. In *International Conference on Learning Representations (ICLR)*. 2025.

C1 Enxin Song*, Wenhao Chai*, Guanhong Wang, Yucheng Zhang, Haoyang Zhou, Feiyang Wu, Haozhe Chi, Xun Guo, Tian Ye, Yanting Zhang, Yan Lu, Jenq-Neng Hwang, and Gaoang Wang. MovieChat: From Dense Token to Sparse Memory for Long Video Understanding. In *Computer Vision and Pattern Recognition (CVPR)*. 2024.

Workshop and Technical Reports

W3 Enxin Song, Wenhao Chai, Weili Xu, Jianwen Xie, Yuxuan Liu, and Gaoang Wang. Video-MMLU: A Massive Multi-Discipline Lecture Understanding Benchmark. In *International Conference on Computer Vision (ICCV) Findings*. 2025.

W2 Weili Xu*, Enxin Song*, Wenhao Chai*, Tian Ye, and Gaoang Wang. Bringing RNNs Back to Efficient Open-Ended Video Understanding. In *Computer Vision and Pattern Recognition (CVPR) Workshop @ Efficient Large Vision Models*. 2025.

W1 Yichen Xu, Zihan Xu, Wenhao Chai, Zhonghan Zhao, Enxin Song, and Gaoang Wang. Devil in the Number: Towards Robust Multi-modality Data Filter. In *International Conference on Computer Vision (ICCV) Workshop @ DataComp*. 2023.

Preprints

P1 Enxin Song, Suhao Yu, Yifei Xu, Barbara Su, Weili Xu, Wenhao Chai, Yao Tang, Jie Deng, Haiyang Xu, and Jiatao Gu. InteractionBench: A Real-Time Interaction Benchmark for Streaming Video Systems. In *arXiv*. 2026.

P2 Enxin Song, Yinuo Xu, Shusheng Yang, Wenhao Chai, and Jiatao Gu. Video-Index: A Curated Meta-Benchmark for Video Understanding. In *arXiv*. 2026.

P3 Shufan Sun, Chen Wang, Enxin Song, Jiatao Gu, and Lingjie Liu. HARMONY: Hierarchical Agentic Reasoning for MONocular Image-to-Scene Synthesis. In *arXiv*. 2026.

P4 Jie Deng, Tiziano Bagnasco, Enxin Song, Tong Nie, Junlin He, Shidong Cao, Alessandro Stocchino, and Wei Ma. SCS-WaveBench: A Simulation-to-Observation Benchmark for Significant Wave Height Forecasting and Imputation in the South China Sea. 2026.

P5 Enxin Song, Wenhao Chai, Xun Guo, Gaoang Wang, Jenq-Neng Hwang, Yan Lu. Fantasy: Transformer Meets Transformer in Text-to-Image Generation. In *OpenReview*. 2024.

Invited Talks

From Seeing to Thinking Lambda AI	Virtual Sept 2025
Video-MMLU: A Massive Multi-Discipline Lecture Understanding Benchmark Workshop on Knowledge MR at ICCV 2025	Oahu, HI Oct 2025
From Compression to Selection: Better and Longer Video Understanding KAUST Rising Stars in AI Symposium	Saudi Arabia Feb 2026

Awards and Honors

KAUST Rising Stars in AI Symposium (Saudi Arabia)	Nov 2025
Outstanding Paper at ICCV 2025 Knowledge-Intensive MR Workshop	Oct 2025
Lambda AI Cloud Credits Grant Sponsorship	Sept 2025
Graduate National Scholarship at Zhejiang University (2.4%)	Sept 2025
Graduate National Scholarship at Zhejiang University (2.4%)	Sept 2024
Undergraduate National Scholarship at Dalian University of Technology (2.3%)	Oct 2021

Professional Service

Workshop Organization

Multimodal Video Agent Workshop on Computer Vision and Pattern Recognition (CVPR) 2025	Nashville, TN June 2025
Long-form Video Understanding Towards Multimodal AI Assistant and Copilot Workshop on Computer Vision and Pattern Recognition (CVPR) 2024	Seattle, WA June 2024

Area Chair

Workshop TAG @ Conference on Robot Learning (CoRL)	2026

Conference and Journal Refereeing

AAAI Conference on Artificial Intelligence (AAAI)	2027
Workshop TAG @ Conference on Robot Learning (CoRL)	2026
International Journal of Computer Vision (IJCV)	2026
International Conference on Machine Learning (ICML)	2026
ACL Rolling Review (ACL ARR)	2026
IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)	2025, 2026
Neural Information Processing Systems (NeurIPS)	2025
International Conference in Learning Representations (ICLR)	2025, 2026, 2027
Computer Vision and Pattern Recognition (CVPR)	2025, 2026
Pattern Recognition and Computer Vision (PRCV)	2023, 2025
IEEE Transactions on Multimedia (TMM)	2024